TFL Component to SAP Traceability Agent

Research question, as written in the proposal

Can an agent link individual components of table, figure, and listing (TFL) shells to the SAP passages that support them, so reviewers can inspect the basis for each reporting specification?

Planning stage. Nothing on these slides has been built, collected, or measured yet.

Background: what we trace, and why it is hard by hand

SAP	Statistical Analysis Plan. Written before unblinding; fixes the analysis populations, endpoints, statistical methods, and how results will be presented.
TFL	Tables, Figures and Listings. The outputs that report a trial: tables summarize, figures plot, listings show one row per participant.
TFL shell	A TFL with its full layout (title, population, rows, columns, axes, footnotes) but no numbers. The shell is the specification of what will be shown.
Traceability	For each shell element, being able to point to the SAP passage(s) that justify it, so a reviewer checks the basis instead of trusting the label.

WHY MANUAL TRACING IS SLOW AND ERROR-PRONE

- One shell row can depend on several SAP sections at once: the population definition, the general analysis conventions, and the endpoint-specific method.
- Shell labels paraphrase the SAP ("LS mean" vs "least squares mean"; "FAS" vs "Full Analysis Set"), and similar analyses exist for different populations.
- A general rule may carry an exception for one endpoint, and a footnote may exist only as a display convention with no SAP passage behind it.

Our objective, and the boundaries we are keeping

WHAT THE AGENT MUST DO (FROM THE PROPOSAL)

01 Take existing TFL shells and the corresponding SAP as input.

02 Split each shell into identifiable components: title and population labels, row and column labels, figure axes, legends and error-bar definitions, listing columns and selection notes, analysis footnotes.

03 For each component, find the SAP passages that support it and check that they apply in that component's analysis context.

04 Assign one of four support statuses and give the rationale.

05 Let a reviewer select a component, inspect the linked quotations, versions and rationale, then accept, correct or reject the link.

06 Export a traceability matrix with reviewer decisions and open questions.

OUTSIDE OUR SCOPE

- Drafting new TFL shells
- Analyzing patient-level data
- Populating results into shells
- Judging whether the statistical methods are appropriate
- Full protocol-to-SAP auditing and SAP amendment tracking (a later extension in the proposal)

CORE SCOPE

Two-arm trials. Continuous-outcome summary tables, mean-by-visit figure shells, simple subject-level endpoint listings. Mapped elements: populations, endpoints, visits, treatment groups, statistics, footnotes.

A running example: one table row, traced to the SAP

Illustrative shell, constructed for this talk

Table 14.2.1 Change from baseline in the continuous primary endpoint at Week 12, Full Analysis Set

Statistic	Treatment A (N = xx)	Treatment B (N = xx)
n	xx	xx
Baseline, mean (SD)	xx.x (xx.xx)	xx.x (xx.xx)
Week 12, mean (SD)	xx.x (xx.xx)	xx.x (xx.xx)
Change from baseline, LS mean (SE)	**xx.x (x.xx)**	**xx.x (x.xx)**

Component C-07 is the highlighted row, in context: FAS, Week 12, continuous primary endpoint, two treatment groups.

Illustrative SAP wording. Fictional, written for this example; not quoted from any real SAP

Section 4.1 Analysis populations

"The Full Analysis Set (FAS) includes all randomized participants who received at least one dose of study treatment."

Section 7.2 Primary endpoint analysis

"Change from baseline at Week 12 will be analyzed with an ANCOVA model including treatment and baseline value. Least squares means and standard errors will be presented by treatment group, using the FAS."

Component	→ SAP passages	→ Verification	→ Status	→ Reviewer
C-07, "Change from baseline, LS mean (SE)"	Sec. 4.1 (population) and Sec. 7.2 (method)	Both apply to Week 12, FAS, primary endpoint; no exception found	supported	accept, or correct / reject

Data requirements: 6 to 8 matched document families

ONE DOCUMENT FAMILY

- One trial's SAP (with version and date) plus the TFL shells written for the same SAP for a two-arm trial.
- Each shell confirmed as belonging to that SAP version, so a link is never checked against the wrong document.
- Recorded for every file: identifier, date, version, and whether the shell is authentic or constructed by us.

THREE CORE SHELL TYPES

Continuous-outcome tables	Summary statistics by treatment group and visit, like Table 14.2.1 in our example.
Mean-by-visit figures	Axes, legends and error-bar definitions must each be traced.
Subject-level listings	Columns and selection notes for each listed participant.

WHAT WE WILL COLLECT AND CONSTRUCT (NOTHING COLLECTED YET)

- Public SAPs with matched shells where available; where authentic shells are missing, constructed shells written to that SAP and labeled "constructed" in the provenance record.
- Challenge cases on purpose: paraphrased labels, similar analyses for different populations, general conventions with endpoint-specific exceptions, unsupported footnotes, incomplete specifications.
- A terminology guide and a short list of approved display conventions, kept separate from SAP evidence.
- No patient-level data and no populated results: the benchmark is about specifications in shells and their documentary support.

Data preparation and annotation: nine steps

01 **Identify matched families**

Confirm which shells belong to which SAP version.

02 **Record provenance**

Identifiers, dates, versions, authentic vs constructed.

03 **Parse shells into components**

Titles, population labels, rows, columns, axes, legends, footnotes.

04 **Assign stable component IDs and locations**

C-07 keeps its ID across runs and exports.

05 **Preserve context**

Endpoint, population, treatment group, visit; disambiguate "n" and "baseline".

06 **Locate SAP passages**

All passages needed, with exact quotations and section / page.

07 **Build the reference set**

Quotation, location, support label, short rationale; several valid passages allowed.

08 **Keep reference apart from retrieval data**

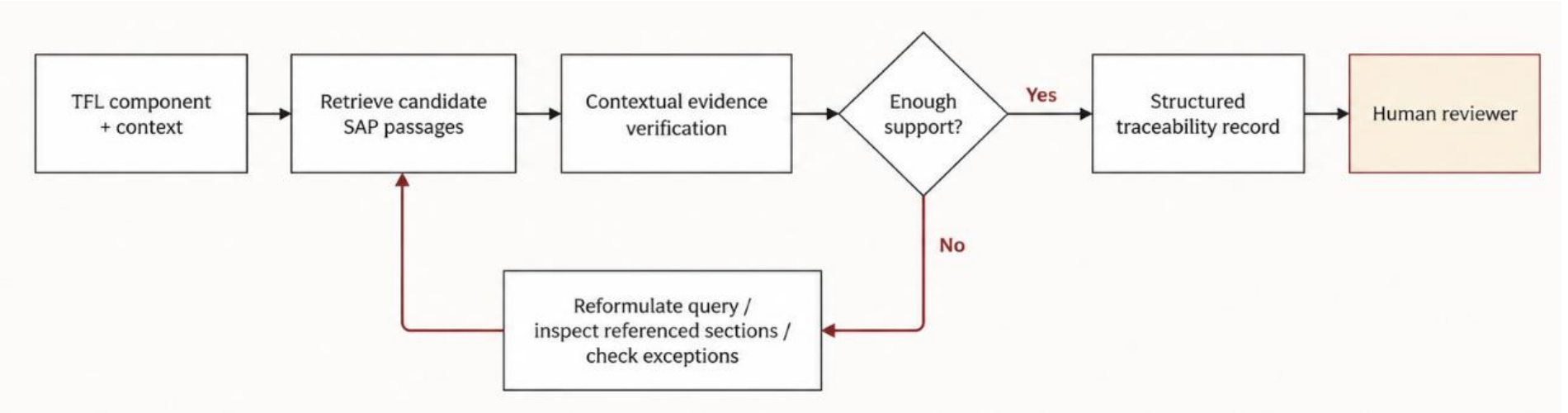
Labels never enter the agent's document collection.

09 **Split by whole family**

Development vs held-out test; a family is never split.

Why this matters: the reference set is the standard every result is scored against; separating it from the agent's retrieval data prevents leakage; splitting by whole family tests generalization to unseen SAPs instead of memorized shells.

Proposed agent workflow



WHAT EACH STAGE DOES

- Parsing: read the supplied SAP and shell versions; turn each shell into selectable components with their context (tool).
- Retrieval: for each component, search definitions, general conventions, endpoint-specific methods and referenced sections (tool). This will be tested with different types of retrieval, ultimately being some form of hybrid retrieval (BM25 + embeddings, potentially reranker)
- Evidence Verification: check that candidate passages apply in this component's context. If not, employ an iterative evidence gathering loop; combine evidence across sections; check exceptions; keep ambiguity when it cannot be resolved (tool + model).
- Structured output: once it has enough support, one record per component with status, quotations, locations and rationale, validated against a schema.
- Reviewer: inspects, then accepts, corrects or rejects; decisions are exported with the matrix.

WHY KEYWORD SIMILARITY IS NOT ENOUGH

- Our C-07 row matches "LS mean" in several SAP sections. Only the one scoped to Week 12, the FAS and the primary endpoint supports it.
- A population label needs two things: the definition in Section 4.1 and the analysis section that applies it (Section 7.2).
- The proposal's rule: a relevant mention alone is not sufficient evidence that a rule applies to the shell.
- The model chooses the next action (parse, retrieve, verify); deterministic tools do the reading and checking; a human makes the final call.

Evidence classification and human review

THE FOUR STATUSES, EXACTLY AS IN THE PROPOSAL

supported

All passages needed for the component were found and apply in its context.

partially supported

Some evidence applies; a qualifier, exception or second passage is missing.

not found in the supplied SAP

No supporting passage in this SAP version. Recorded as missing, never as proof of a violation.

needs review

Competing interpretations, contradictions, or ambiguity the agent should not resolve alone.

EXAMPLE TRACEABILITY RECORD (ILLUSTRATIVE)

```
component_id: C-07
shell: Table 14.2.1 (shell v1.0)   sap: SAP v2.0, 2024-03-01
component: "Change from baseline, LS mean (SE)"
context: FAS | Week 12 | primary endpoint | two arms
evidence:
  - Sec 4.1, p.12: "The Full Analysis Set (FAS) includes ..."
  - Sec 7.2, p.31: "... Least squares means and standard
    errors will be presented by treatment group, using the FAS."
support_status: supported
rationale: population and method both scoped to this endpoint
display_convention: none
reviewer_decision: pending
```

WHAT THE REVIEWER DOES

- Selects a component and reads the linked quotations, locations and SAP version in a source panel.
- Accepts the link, corrects it (different passage or status), or rejects it; each decision is recorded.
- Exports the matrix together with unresolved questions.
- Display conventions (for example a standard footnote on how percentages are computed) are labeled as conventions, never given a manufactured SAP link.

How we plan to evaluate, and what will be hard

EVALUATION DESIGN: THREE LEVELS

Evidence Retrieval: finding the right passages

- Component identification evaluated separately to isolate parsing errors.
- Recall@k and link precision/recall, including multi-passage evidence and equivalent spans.

Contextual Verification: how well evidence actually supports component

- Evidence-link precision: proportion of proposed links that genuinely substantiate the component.
- Citation accuracy and support-status accuracy against reviewed reference labels.
- Correct abstention when evidence is missing or ambiguous.
- Challenge cases: population/endpoint mismatches, exceptions, paraphrases, and convention-only components.

End-to-End Usefulness:

- Reviewer correction rate and time required to reach trustworthy links.

EXPECTED CHALLENGES

- Similar population definitions (FAS vs Safety Set differing by one clause).
- Paraphrased labels: "LS mean" vs "least squares mean", "CFB" vs "change from baseline".
- Evidence scattered across sections that must be combined.
- Conditional rules: a general convention with an endpoint-specific exception.
- Ambiguous footnotes: convention, SAP rule, or neither?
- Too few authentic shells, so constructed shells may be needed and must be flagged.
- Annotation disagreement among us; we will need a written guide and double review.

Team responsibilities, next steps, and questions for the TA

01

Data acquisition and provenance

Find SAPs and shells, confirm pairings, record versions and authenticity, build the manifest.

02

Annotation and reference set

Component inventory, stable IDs, context, SAP passages, support labels, rationale; terminology guide.

03

Agent pipeline and reviewer interface

Parsing, retrieval and verification tools; schema; source panel with accept / correct / reject; export.

04

Validation, evaluation, documentation

Metrics, baselines, held-out runs, failure-case review, reproducible code and report.

Modular allocation: with 3 members, one member combines data acquisition and annotation (01 + 02). Every member reviews the schema, the status definitions and the annotation guide, and every member annotates part of the reference set.

IMMEDIATE NEXT STEPS

01 Confirm group size and lock the allocation above.

02 Search public sources for candidate SAP and shell families; list what is authentic vs what we must construct.

03 Draft the component schema and the annotation guide; annotate one family together as calibration.

04 Freeze the development / held-out split before any agent work starts.

QUESTIONS FOR THE TA

01 If authentic shells are scarce, what mix of authentic and constructed shells is acceptable for the 6 to 8 families?

02 For the reference set, is single-annotator labeling with a second-person check enough, or should every component be double-annotated?

03 For the final demo, is a lightweight reviewer interface (a notebook or simple web page) sufficient, or is a standalone application expected?